Well, that's the rough idea, but it's worthwhile sharpening the description a bit. The horizon problem stems from the fact that like a ball tossed upward, the dragging pull of gravity causes the expansion rate of the universe to slow down. This means that, for example, to halve the separation between two locations in the cosmos we must run the film back more than halfway toward its beginning. In turn, we see that to halve the separation we must more than halve the time since the big bang. Less time since the bang—proportionally speaking—means it is harder for the two regions to communicate, even though they get closer.

Guth's resolution of the horizon problem is now simple to state. He found another solution to Einstein's equations in which the very early universe undergoes a brief period of enormously fast expansion—a period during which, in fact, it "inflates" in size at an unheralded exponential expansion rate. Unlike the case of a ball that slows down after being tossed upward, exponential expansion gets faster as it proceeds. When we run the cosmic film in reverse, rapid accelerating expansion turns into rapid decelerating contraction. This means that to halve the separation between two locations in the cosmos (during the exponential epoch) we need run the film back less than halfway—much less, in fact. Running the film back less implies that the two regions will have had more time to communicate thermally and, like hot soup and air, they will have had ample time to come to the same temperature.

Through Guth's discovery and later important refinements made by André Linde, now of Stanford University, Paul Steinhardt and Andreas Albrecht, then of the University of Pennsylvania, and many others, the standard cosmological model was revamped into the inflationary cosmological model. In this framework, the standard cosmological model is modified during a tiny window of time—around 10-36 to 10-34 seconds ATB—in which the universe expanded by a colossal factor of at least 1030, compared with a factor of about a hundred during the same time interval in the standard scenario. This means that in a brief flicker of time, about a trillionth of a trillionth of a trillionth of a second ATB, the size of the universe increased by a greater percentage than it has in the 15 billion years since. Before this expansion, matter that is now in far-flung regions of the cosmos was much closer